

PAYERNE, Switzerland

WMO: 066100

Lat: 46.8117N

Lon: 6.9425E

Elev: 491

StdP: 95.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.1	(c) -6.1	(d) -11.0	(e) 1.5	(f) -6.7	(g) -8.9	(h) 1.9	(i) -4.6	(j) 9.9	(k) 8.0	(l) 8.9	(m) 6.7	(n) 2.2	(o) 50	(p) 0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.2	(c) 30.3	(d) 19.8	(e) 28.4	(f) 19.2	(g) 26.6	(h) 18.6	(i) 20.7	(j) 28.0	(k) 19.9	(l) 26.6	(m) 19.1	(n) 25.2	(o) 2.6	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.9	23.4	17.5	13.3	22.5	16.9	12.8	21.7	61.8	28.2	59.0	26.7	56.3	25.4	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.5	(c) 5.6	(e) -11.0	(f) 33.0	(g) 2.6	(h) 2.1	(i) -13.0	(j) 34.5	(k) -14.5	(l) 35.7	(m) -16.0	(n) 36.9	(o) -17.9	(p) 38.5		
(a) 7.6	(b) 6.5	(c) 5.6	(e) -11.4	(f) 22.3	(g) 2.7	(h) 0.8	(i) -13.4	(j) 22.9	(k) -14.9	(l) 23.3	(m) -16.4	(n) 23.8	(o) -18.4	(p) 24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	0.9	2.1	6.0	9.5	13.7	17.6	19.5	19.0	14.8	10.6	5.0	1.6		
	DBStd	7.50	3.99	4.02	3.69	3.50	3.31	3.55	3.12	3.09	3.19	3.44	3.57	3.83		
	HDD10.0	1146	282	223	131	52	7	1	0	0	3	35	153	260		
	HDD18.3	3170	540	456	383	266	146	57	23	29	113	241	400	518		
	CDD10.0	1171	0	1	7	36	123	227	296	281	146	52	3	0		
	CDD18.3	154	0	0	0	0	3	33	61	51	6	0	0	0		
	CDH23.3	1383	0	0	0	3	49	306	547	431	47	1	0	0		
	CDH26.7	383	0	0	0	0	6	77	167	129	3	0	0	0		
(14) Wind	WSAvg	2.0	2.2	2.3	2.4	2.1	2.0	1.9	1.8	1.6	1.7	1.6	1.9	2.2		
(15) Precipitation	PrecAvg	868	59	52	53	63	85	93	92	91	73	68	78	65		
	PrecMax	1282	150	151	126	160	155	187	236	285	179	140	182	176		
	PrecMin	581	5	1	6	13	33	23	21	15	10	4	4	0		
	PrecStd	166	36	33	31	37	30	45	48	50	47	38	45	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.7	18.8	23.4	27.3	31.6	33.2	32.8	27.0	21.8	15.9	11.9		
		MCWB	8.6	8.8	10.9	14.6	17.5	20.6	20.0	20.0	18.1	15.1	11.7	8.6		
	2%	DB	10.1	11.2	16.4	20.9	24.6	28.9	30.5	29.9	24.7	19.2	13.2	10.2		
		MCWB	7.6	7.4	10.0	12.9	16.2	19.4	19.7	19.7	17.7	14.6	10.4	7.9		
	5%	DB	8.2	9.5	14.2	18.7	22.7	26.8	28.4	27.8	22.6	17.3	11.4	8.7		
		MCWB	6.2	6.5	9.1	11.9	15.2	18.9	19.2	19.2	16.6	13.8	9.3	6.7		
	10%	DB	6.7	7.8	12.2	16.5	20.5	24.8	26.4	25.7	20.6	15.7	10.0	7.3		
		MCWB	4.9	5.5	8.1	10.8	14.4	17.8	18.4	18.5	15.9	13.1	8.3	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.1	11.9	15.0	18.9	22.0	21.7	21.5	19.6	16.4	12.2	9.4		
		MCDB	11.4	12.5	17.3	22.0	25.6	29.6	29.6	28.9	25.0	19.7	14.9	11.0		
	2%	WB	7.8	7.7	10.6	13.4	17.2	20.4	20.6	20.5	18.3	15.3	10.9	8.2		
		MCDB	9.7	10.7	15.2	19.6	22.9	27.0	28.2	27.8	23.2	18.0	12.8	9.8		
	5%	WB	6.3	6.7	9.5	12.3	16.0	19.4	19.7	19.7	17.3	14.3	9.6	7.0		
		MCDB	8.2	9.2	13.5	17.7	21.1	25.3	26.8	26.2	21.4	16.7	11.1	8.5		
	10%	WB	5.0	5.5	8.4	11.2	14.8	18.3	18.9	18.9	16.3	13.3	8.5	5.7		
		MCDB	6.5	7.6	11.9	15.8	19.5	23.8	25.1	24.6	19.7	15.3	9.7	7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.7	9.1	10.2	10.1	10.6	11.2	10.7	9.6	7.7	5.5	4.7		
		MCDBR	6.9	9.7	12.8	14.7	14.1	14.0	14.9	14.6	12.5	10.2	7.8	6.4		
	5% WB	MCWBR	5.0	6.5	7.4	7.7	6.9	6.3	5.8	5.8	6.3	6.1	5.5	4.7		
		MCWBR	6.6	9.0	11.6	13.5	12.6	13.0	13.4	13.0	11.4	9.2	7.3	5.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.315	0.353	0.392	0.403	0.404	0.403	0.394	0.373	0.351	0.317	0.288		
		taud	2.426	2.365	2.329	2.260	2.277	2.316	2.312	2.349	2.390	2.438	2.472	2.446		
	Ebn at Noon	Ebn at Noon	808	853	867	862	863	862	857	851	837	801	767	766		
		Edh at Noon	68	89	107	126	129	125	124	115	101	83	66	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.28	2.14	3.30	4.36	5.00	5.93	5.79	5.04	3.92	2.51	1.39	1.04		
	RadStd	RadStd	0.15	0.32	0.48	0.63	0.45	0.56	0.50	0.46	0.35	0.25	0.18	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.87	+0.37	+0.39	N/A	N/A	N/A	N/A
(47) Regional (107 neighbors)		N/A	N/A	N/A	+0.74	+0.40	+0.30	N/A	N/A	+57	+18

Nomenclature: See separate page